

Auto-deriving Tech Status from the fault + part — proposal

Grounded in the TM's PMCS 'Not Fully Mission Capable If' criteria. Required & filled before the sheet generates. Decide before we build.

HOW TECH STATUS IS ACTUALLY DECIDED

A fault either deadlines the vehicle or it doesn't. The TM's PMCS 'NOT FULLY MISSION CAPABLE IF' column is the authority — it states exactly which faults make the vehicle non-mission-capable.



Since this is a PARTS request (waiting on supply), a deadline fault most often → NMCS.

THE DATA IS IN YOUR CORPUS

'Not Fully Mission Capable If'	381 pages
PMCS tables	1,158 pages
'mission capable'	439 pages

(sample index of 45k pages — the full index is ~40x larger)

SIGNAL SOURCES (combine for confidence)

A · PMCS-grounded (authoritative)

Search the vehicle's PMCS pages for the fault/part terms; read the matched 'Not Fully Mission Capable If' line and cite it verbatim.

Doctrinally correct + cited. Coverage depends on OCR (partial now, grows).

B · Keyword rules (works today)

Offline deadline-term classifier: e.g. no-start / steering / brakes / Class III leak → likely NMC; gauge / lamp / cosmetic → PMC-FMC. Extensible JSON.

Instant, offline, no OCR needed. Approximate — a starting guess only.

C · Learned from history

Same fault + part on past sessions → the tech status the mechanic confirmed before. Improves with use (like the parts learning).

Gets smarter over time. Empty on a fresh install.

RECOMMENDED PIPELINE · propose with evidence → mechanic confirms (mandatory) → onto the sheet



WHY A CONFIRM GATE (not silent auto-fill)

Tech status is a formal readiness call with real consequences. The app should PROPOSE it with cited evidence and require a human to confirm — never silently decide, and never invent a deadline. It stays a faithful, cited determination (the project's safety stance).

It is still MANDATORY: the sheet will not generate with Tech Status blank, so it's always present by the end — just human-confirmed.

RECOMMENDATION

- Hybrid: A (PMCS, cited) → B (rules) → C (history), best evidence wins.
- Suggest + mandatory confirm at 'Generate sheet' — pre-filled, editable, blocks if blank.
- Use full codes (FMC / PMCM / PMCS / NMCM / NMCS); parts-request defaults toward the supply (S) code.
- Ship the keyword-rule pack first (works now); PMCS-cited evidence strengthens as OCR completes.